

Chapter 19

Energy and momentum in media

19.1 Maxwell's equations with materials and sources

In the presence of material and sources, the Maxwell equations read

$$\vec{\nabla} \times \vec{E} = -\frac{1}{c} \frac{\partial \vec{B}}{\partial t} \quad , \quad \vec{\nabla} \cdot \vec{B} = 0 \quad (19.1)$$

$$\vec{\nabla} \cdot \vec{D} = 4\pi \rho \quad , \quad \vec{\nabla} \times \vec{H} = \frac{4\pi}{c} \vec{j} + \frac{1}{c} \frac{\partial \vec{D}}{\partial t} \quad (19.2)$$

where ρ and $\vec{j}$ are the *free* charge and current densities (note that it is $\vec{D}$, not $\vec{E}$, that appears in the displacement term, since this is the $\vec{H}$ equation). We supplement these by the constitutive relations

$$\vec{D} = \epsilon \vec{E} \quad , \quad \vec{B} = \mu \vec{H} \quad (19.3)$$

For dynamical conductors, we add

$$\vec{j} = \sigma \vec{E} \quad (19.4)$$

with σ being the conductivity. The free charge and current densities satisfy the conservation statement

$$\vec{\nabla} \cdot \vec{j} + \frac{\partial \rho}{\partial t} = 0 \quad (19.5)$$

19.2 The Poynting theorem in materials

The force per unit volume on the sources is

$$\vec{f} = \rho \vec{E} + \frac{\vec{j}}{c} \times \vec{B} \quad (19.6)$$

– the volume versions of the single-charge forces $q \vec{E}$ and $q \vec{v} \times \vec{B}/c$. The energy density of the fields in the material is

$$u = \frac{1}{8\pi} \left(\vec{E} \cdot \vec{D} + \vec{H} \cdot \vec{B} \right) \quad (19.7)$$

To find the Poynting theorem in the material, take the time derivative (using linearity of the constitutive relations) and use the Maxwell equations

$$\begin{aligned} -\frac{\partial u}{\partial t} &= -\frac{1}{4\pi} \vec{E} \cdot \frac{\partial \vec{D}}{\partial t} - \frac{1}{4\pi} \vec{H} \cdot \frac{\partial \vec{B}}{\partial t} \\ &= \vec{\nabla} \cdot \left[\frac{c}{4\pi} \left(\vec{E} \times \vec{H} \right) \right] + \vec{E} \cdot \vec{j} \end{aligned} \quad (19.8)$$

where we identify the Poynting vector in the material

$$\vec{S} = \frac{c}{4\pi} \vec{E} \times \vec{H} \quad (19.9)$$

Integrating over a volume, with U the field energy,

$$-\frac{dU}{dt} = \oint \vec{S} \cdot d\vec{a} + \int \vec{j} \cdot \vec{E} dV \quad (19.10)$$

What is the last term? It is the energy drain to the sources

$$\int \vec{j} \cdot \vec{E} dV = \frac{d}{dt} U_{\text{mech}} \quad (19.11)$$

so that

$$\frac{d}{dt} (U_{\text{EM}} + U_{\text{mech}}) = - \oint \vec{S} \cdot d\vec{a} \quad (19.12)$$

19.3 Momentum conservation and the stress tensor

Consider the total force on the sources, obtained by integrating (19.6) and eliminating ρ and $\vec{j}$ using (19.2)

$$\begin{aligned}\vec{F} = \frac{d\vec{P}_{\text{mech}}}{dt} &= \int \left(\rho \vec{E} + \frac{\vec{j}}{c} \times \vec{B} \right) dV \\ &= \frac{1}{4\pi} \int dV \left[\vec{E} (\vec{\nabla} \cdot \vec{D}) + (\vec{\nabla} \cdot \vec{B}) \vec{H} \right. \\ &\quad \left. + (\vec{\nabla} \times \vec{H}) \times \vec{B} - \frac{1}{c} \frac{\partial \vec{D}}{\partial t} \times \vec{B} \right] \quad (19.13)\end{aligned}$$

where there is no cost to adding the $(\vec{\nabla} \cdot \vec{B}) \vec{H}$ term since $\vec{\nabla} \cdot \vec{B} = 0$. For the last term, write

$$\begin{aligned}-\frac{1}{c} \frac{\partial \vec{D}}{\partial t} \times \vec{B} &= -\frac{1}{c} \frac{\partial}{\partial t} (\vec{D} \times \vec{B}) + \frac{1}{c} \vec{D} \times \frac{\partial \vec{B}}{\partial t} \\ &= -\frac{1}{c} \frac{\partial}{\partial t} (\vec{D} \times \vec{B}) - \vec{D} \times (\vec{\nabla} \times \vec{E}) \quad (19.14)\end{aligned}$$

using Faraday's law in the last step. Moving the total time derivative to the left-hand side, we identify the electromagnetic momentum in the material

$$\vec{P}_{\text{EM}} = \frac{1}{4\pi c} \int \vec{D} \times \vec{B} dV \quad (19.15)$$

and we get

$$\begin{aligned}\frac{d\vec{P}_{\text{mech}}}{dt} + \frac{d\vec{P}_{\text{EM}}}{dt} &= \frac{1}{4\pi} \int dV \left[(\vec{\nabla} \cdot \vec{D}) \vec{E} + (\vec{\nabla} \cdot \vec{B}) \vec{H} \right. \\ &\quad \left. + (\vec{\nabla} \times \vec{H}) \times \vec{B} + (\vec{\nabla} \times \vec{E}) \times \vec{D} \right] \\ &= \frac{1}{c} \int \partial_i T_{\text{EM}}^{ij} dV \quad (19.16)\end{aligned}$$

where the stress tensor, generalized to materials, is

$$T_{\text{EM}}^{ij} = \frac{c}{4\pi} \begin{pmatrix} H_x B_x + D_x E_x & H_x B_y + D_x E_y & H_x B_z + D_x E_z \\ -\frac{1}{2} \left(\vec{E} \cdot \vec{D} + \vec{B} \cdot \vec{H} \right) & H_y B_y + D_y E_y & H_y B_z + D_y E_z \\ H_x B_y + D_x E_y & -\frac{1}{2} \left(\vec{E} \cdot \vec{D} + \vec{B} \cdot \vec{H} \right) & H_y B_z + D_y E_z \\ H_x B_z + D_x E_z & H_y B_z + D_y E_z & -\frac{1}{2} \left(\vec{E} \cdot \vec{D} + \vec{B} \cdot \vec{H} \right) \end{pmatrix} \quad (19.17)$$

The interpretation: T_{EM}^{ij} is c times the flux of the i th component of momentum through a surface oriented along the j direction. The force on a surface element of area Δa with unit normal $\hat{n}$ is then

$$\Delta F^i = -\frac{1}{c} T_{\text{EM}}^{ij} n^j \Delta a \quad (19.18)$$

so that

$$\Delta \vec{F} \cdot \hat{n} = -\frac{1}{c} T_{\text{EM}}^{ij} n^i n^j \Delta a \Rightarrow \text{pressure on the surrounding} = -\frac{1}{c} T_{\text{EM}}^{ij} n^i n^j \quad (19.19)$$